

Research Methods — Module 08: Planning — Study Questions

1. List the five main research frontiers for Active Inference. Which is most promising and why?
2. How does the scale problem (neurons to societies) relate to the renormalization group from the Advanced Theory track?
3. What advantages might Active Inference offer over deep RL for building safe AI?
4. What additional constraints beyond Active Inference might be needed for consciousness?
5. How does the FEP apply to morphogenesis? What does "free energy minimization" mean for a developing embryo?
6. How can Active Inference inform climate adaptation strategies?
7. What are the components of a strong Active Inference research proposal?
8. How do you demonstrate innovation in an Active Inference grant proposal?
9. What preliminary data strengthens a proposal? How do you generate it?
10. How do the four tracks of the 401 course (philosophy, neuroscience, theory, methods) integrate?
11. What does philosophy contribute that mathematics alone cannot?
12. What does neuroscience contribute that theory alone cannot?
13. What does the methods track contribute that the other tracks cannot?
14. Name three key Active Inference labs and their research focus.
15. What software tools should every Active Inference researcher know?
16. What are the most important unresolved theoretical debates in Active Inference?
17. How might Active Inference integrate with deep learning architectures?
18. What is the most important methodological challenge facing Active Inference research?
19. How should Active Inference engage with its critics?
20. Write a personal research statement: What Active Inference problem would you study, and how?